

March 1993
Edition 1.0

FUJITSU

DATA SHEET

MBM27C4001-12-X/-15-X

CMOS 4M-BIT UV EPROM

CMOS 4,194,304-BIT UV ERASABLE READ ONLY MEMORY

The Fujitsu MBM27C4001 EPROM is a high speed read-only memory that is UV-erasable and electrically programmable. The device contains 4,194,304 programmable or reprogrammable bits organized in a 524,288-byte/8-bit format. The MBM27C4001 is housed in a 32-pin DIP with a transparent lid; when the lid is properly exposed to an ultraviolet light source, a previously programmed bit pattern is erased in approximately 15 to 20 minutes. A new bit pattern can then be written into memory.

The MBM27C4001 EPROM is fabricated using CMOS double poly-silicon gate technology with stacked single-transistor gate cells. The MBM27C4001 is an excellent choice for system development work and in other applications where program changes are frequently necessary. Once programmed, the device requires only a single +5V power supply; the current requirements are exceptionally low in both the active and standby modes of operation.

- 524,288-byte/8-bit organization with on-chip decoding
- Single-byte or Four-byte programming capability with high-speed algorithms
- Static operation (no clocks required)
- Fast access time:

MBM27C4001-12-X = 120ns (max)

MBM27C4001-15-X = 150ns (max)

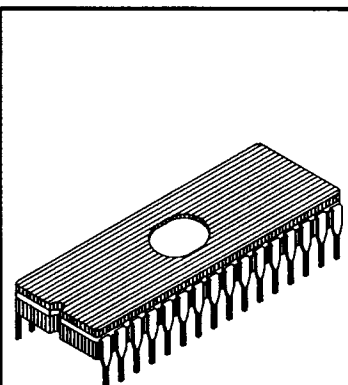
- Easy and simple memory expansion via @ pin
- Three-state output for wired-OR capability
- TTL-compatible inputs/outputs
- Single =5V ($\pm 10\%$) power supply with low current drain:

Active operation = 50mA (max)

Standby operation = 1.0mA(max)-TTL Level

Standby operation = 100 μ A(max)-CMOS Level

- Fast programming: 0.1ms pulse
- Programming voltage: +12.5V
- JEDEC approved 32-pin Ceramic DIP(Suffix: Z)



CERAMIC PACKAGE
DIP-32C-C02

PIN ASSIGNMENT

(TOP VIEW)

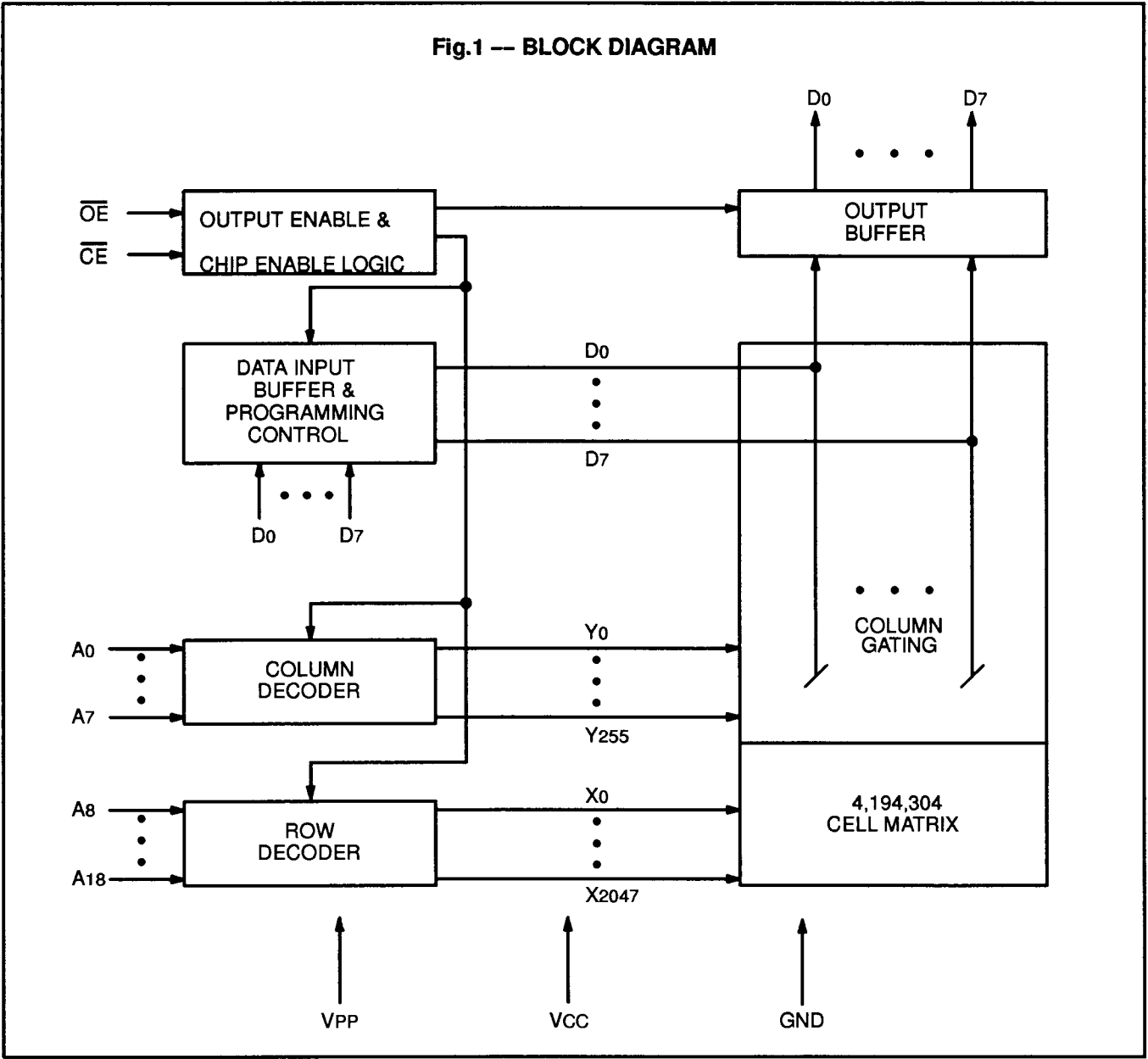
VPP	1	32	VCC
A16	2	31	A18
A15	3	30	A17
A12	4	29	A14
A7	5	28	A13
A6	6	27	A8
A5	7	26	A9
A4	8	25	A11
A3	9	24	OE
A2	10	23	A10
A1	11	22	CE
A0	12	21	D7
D0	13	20	D6
D1	14	19	D5
D2	15	18	D4
GND	16	17	D3

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields. However, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

ABSOLUTE MAXIMUM RATINGS (see NOTE)

Rating	Symbol	Value	Unit
Supply Voltage with respect to GND	VCC	-0.6 to +7.0	V
Programming Voltage with respect to GND	VPP	-0.6 to +14.0	V
Input/Output Voltage (except for A9 with respect to GND)	VIN1, VOUT	-0.6 to VCC+0.6	V
Programming Voltage with respect to GND	VIN2	-0.6 to +13.5	V
Temperature under Bias	TBIAS	-50 to +95	°C
Storage Temperature Range	TSTG	-65 to +125	°C

NOTE: Permanent device damage may occur if the above **Absolute Maximum Ratings** are exceeded. Functional operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.



CAPACITANCE ($T_A = 25^\circ \text{C}$, $f = 1\text{MHz}$)

Parameter	Symbol	Values			Unit
		Min	Typ	Max	
Input Capacitance ($V_{IN}=0V$)	CIN		10	13	pF
Output Capacitance ($V_{OUT}=0V$)	COUT		7	10	pF

PIN DESCRIPTION

Symbol	Pin No.	Function
VPP	1	+12.5V programming voltage
$\overline{OE}$	24	Output enable. When $\overline{OE}$ and $\overline{CE}$ are active Low strobe is active High; all output lines(D0 to D7) are enabled.
A0 to A18	2 to 12, 23 to 31	Address lines
D0 to D7	13 to 15, 17 to 21	Three-state output data lines
GND	16	Circuit ground
$\overline{CE}$	22	When active Low, the device is enabled for data read.
VCC	32	+5V power supply

FUNCTIONS AND PIN CONNECTIONS

1. Read Mode

Mode \ Symbol	A0 to A18	D0 to D7	$\overline{OE}$	$\overline{CE}$	VCC	VPP	GND
Standby	X	Hi-Z	X	VIH	+5V	+5V	GND
Read	AIN	DOUT	VIL	VIL	+5V	+5V	GND
Output Disable	AIN	Hi-Z	VIH	VIL	+5V	+5V	GND
Electronic Signature	A0	CODE	VIL	VIL	+5V	+5V	GND

Legend: X="H" or "L"

2. Single-Byte Programming Mode

Mode \ Symbol	A0 to A18	D0 to D7	$\overline{OE}$	$\overline{CE}$	VCC	VPP	GND
Data Latch	AIN	DIN	VIL	VIH	+6V	+12.5V	GND
Program	AIN	Hi-Z	VIH	VIL	+6V	+12.5V	GND
Verify	AIN	DOUT	VIL	VIL	+6V	+12.5V	GND
Program Inhibit	X	Hi-Z	VIH	VIH	+6V	+12.5V	GND

Legend: X="H" or "L"

3. Four-Byte Programming Mode

Mode \ Symbol	A0 , A1	A2 to A18	D0 to D7	$\overline{OE}$	$\overline{CE}$	VCC	VPP	GND
Data Latch	AIN	AIN	DIN	VIL	VIH	+6V	+12.5V	GND
Program	X	AIN	Hi-Z	VIH	VIL	+6V	+12.5V	GND
Verify	AIN	AIN	DOUT	VIL	VIL	+6V	+12.5V	GND
Program Inhibit	X	X	Hi-Z	VIH	VIH	+6V	+12.5V	GND

Legend: X='H' or 'L'

RECOMMENDED OPERATING CONDITIONS

(Referenced to GND)

Parameter	Symbol	Min	Typ	Max	Unit
Supply Voltage	VCC	4.5	5.0	5.5	V
Supply Voltage *1	VPP	VCC -0.6	VCC	VCC +0.6	V
Input High Level	VIH	2.0		VCC +0.3	V
Input Low Level	VIL	-0.3		0.8	V
Operating Temperature	TA	-40		85	°C

*1 : VPP supply voltage is applied posterior to or coincident with VCC supply voltage and cut off prior to or coincident with VCC supply voltage.

DC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)

Parameter	Symbol	Conditions	Values			Unit
			Min	Typ	Max	
Input Leakage Current	ILI	VIN = VCC = 5.5V			10	μA
Output Leakage Current	ILO	VOUT = VCC = 5.5V			10	μA
VCC Standby Current	ISB1	$\overline{CE}$ = VIH			1.0	mA
VCC Standby Current	ISB2	$\overline{CE}$ = VCC ±0.3V			100	μA
VCC Operation Current	ICC	Cycle = min., IOUT = 0mA			50	mA
VPP Supply Current	IPP	VPP = VCC ±0.6V			100	μA
Output High Level	VOH1	IOH = -400μA	2.4			V
Output High Level	VOH2	IOH = -100μA	VCC -0.7			V
Output Low Level	VOL	IOL = 2.1mA			0.45	V

AC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)

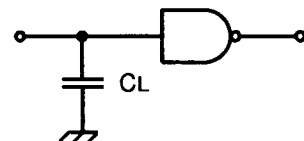
Parameter	Symbol	MBM27C4001-12-X Values		MBM27C4001-15-X Values		Unit
		Min	Max	Min	Max	
Address Access Time	tACC		120		150	ns
$\overline{CE}$ to Output Delay Time	tCE		120		150	ns
$\overline{OE}$ to Output Delay Time	tOE		70		70	ns
$\overline{CE}$ or $\overline{OE}$ to Output Float Delay (Note)	tDF		60		60	ns
Address to Output Hold Time	tOH	0		0		ns
$\overline{CE}$ or $\overline{OE}$ to Output Active	tDV	0		0		ns

NOTE: Output Float is defined as the point where data is no longer driven.

AC TEST CONDITIONS

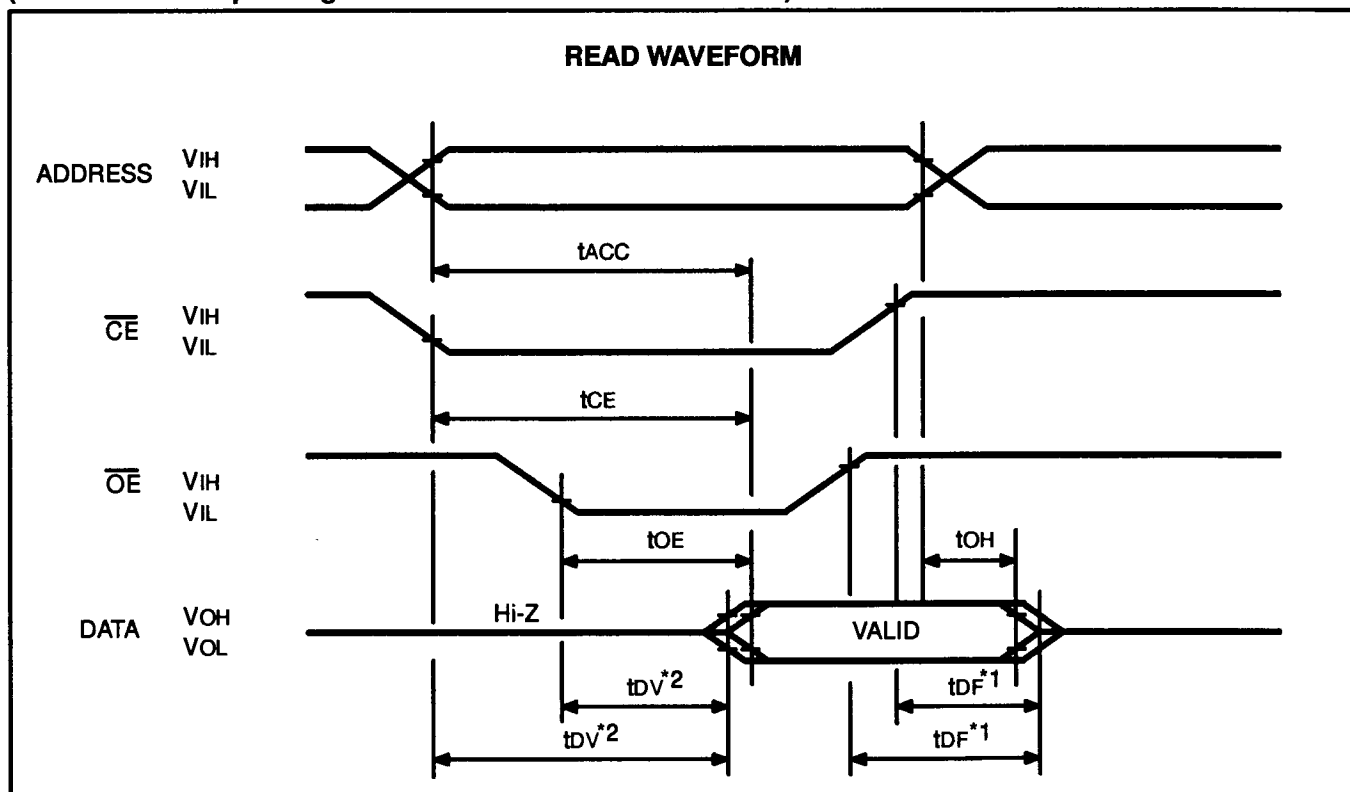
Fig. 2 —AC TEST CONDITIONS(INCLUDING PROGRAMMING)

Input Pulse Levels: 0.45V to 2.4V
 Input Rise/Fall Times: $\leq 20\text{ns}$
 Input Reference Levels: 0.8V to 2.0V
 Output Reference Levels: 0.8V to 2.0V
 Output Load: 1 TTL gate and $C_L = 100\text{pF}$



AC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)



NOTE: *1: t_{DF} is specified by either of $\overline{CE}$ or $\overline{OE}$ changing to high earlier.

*2: t_{DV} is specified by either of $\overline{CE}$ or $\overline{OE}$ changing to low later.

PROGRAMMING / ERASING INFORMATION

PROGRAMMING

Single-Byte Programming. When +12.5V(± 0.3) volts are applied to Vpp, +6(± 0.25) volts are to VCC and $\overline{CE}$, $\overline{OE} = V_{IH}$ (TTL), the programming mode is initiated. Address is selected for programming by address pins A0 to A18 and input data is applied to output pins D0 to D7. When both address and data are stable, a 0.1-millisecond negative pulse is applied to the $\overline{OE}$ pin. Upon verification of written data should be applied to complete the programming of one byte. Refer to the PROGRAMMING FLOWCHART that follows for step-by-step programming procedures.

Four-Byte Programming. Compared to the Single-byte programming, the Four-byte programming method reduces programming time by about 75%. Voltages applied to VPP and VCC are the same as in the Single-byte programming; however some logic levels differ (refer to "Four Byte Programming" in the truth table). In conjunction with $\overline{OE}$ pin, address pins A0 and A1 are used to latch four bytes of data. When both address and data are stable, a 0.1-millisecond negative pulse is applied to the $\overline{OE}$ pin. Upon verification of written data should be applied to complete the programming of four-byte.

Caution

In program mode (VPP=12.5V(± 0.3)), a continue us TTL low-level Voltage should not be applied to the $\overline{CE}$. Also a 0.1-microfarad capacitor must be connected between VPP and ground to prevent excessive voltage transient. Neglecting

either of these precautions may cause device failure.

Electronic Signature/Programming Algorithm. When MBM27C4001 is shipped from the factory, all memory cells (4,194,304 bits) are set to the High state (logic 1). During the programming procedure, affected bit cells are set to the Low (logic 0) state.

Manufacturer and device codes are electronically stored in each device. The Electronic Signature Code List is shown in the table preceding the ELECTRICAL CHARACTERISTICS.

ERASING

In order to clear all memory cells of programmed contents, the MBM27C4001 must be exposed to an ultraviolet light source. To completely erase the memory (restore all cells to a logic 1 state), a dosage of 10Wsec/cm² is required. The required exposure can be obtained by using a UV-lamp with a wavelength of 2537 Angstroms and with an intensity of 12mW/cm². Remove all filters from the lamp and clean the transparent lid of the MBM27C4001 with a non-abrasive cleaner. Hold the MBM27C4001 approximately one inch from the light source for 15 to 20 minutes. (Note. The MBM27C4001 and other similar devices can be erased by light sources with longer wavelengths; however, the erasing time is much greater. Nonetheless, exposure to fluorescents or sunlight will severely degrade and eventually erase the memory. When used in a lighted environment, it is recommended that the transparent window be covered with an opaque label.)

ELECTRONIC SIGNATURE CODE LIST

Definition	A0	A1 to A8	A9	A10 to A18	$\overline{OE}$	$\overline{CE}$	D0	D1	D2	D3	D4	D5	D6	D7	HEX
Manufacture	VIL	VIL	12(± 0.5)V	VIL	VIL	VIL	0	0	1	0	0	0	0	0	#04
Device	VIH	VIL	12(± 0.5)V	VIL	VIL	VIL	0	0	1	0	1	1	1	1	#F4

PROGRAMMING / ERASING INFORMATION (Cont'd)

DC CHARACTERISTICS

(TA = 25 °C ± 5°C, VCC^{*1} = 6V ± 0.25V, VPP^{*2} = 12.5V ± 0.3V)

Parameter	Symbol	Conditions	Values			Unit
			Min	Typ	Max	
Input Leakage Current	I _{LI}	V _{IN} = 6.25V/0V			10	μA
VPP Supply Current (Single-Byte)	I _{PP1}	$\overline{CE} = V_{IL}, \overline{OE} = V_{IH}$			30	mA
VPP Supply Current (Four-Byte)	I _{PP2}	$\overline{CE} = V_{IL}, \overline{OE} = V_{IH}$			50	mA
VPP Supply Current (Inhibit)	I _{PP3}	$\overline{CE} = V_{IH}, \overline{OE} = V_{IH}$			5	mA
VPP Supply Current (Verify)	I _{PP4}	$\overline{CE} = V_{IL}, \overline{OE} = V_{IL}$			5	mA
VCC Supply Current	I _{CC}				50	mA
Input Low Level	V _{IL}		-0.1		0.8	V
Input High Level	V _{IH}		2.0		VCC + 0.3	V
Output Low Level	V _{OL}	I _{OL} = 2.1mA			0.45	V
Output High Level	V _{OH}	I _{OH} = -400μA	2.4			V

NOTE : *1 VCC must be applied either coincidentally or before VPP and removed either coincidentally or after VPP.

*2 VPP must not be 13.5volts or more including overshoot. Permanent device damage may occur if the device is taken out or put into socket remaining VPP=12.5 volts. Also, during $\overline{CE}=V_{IL}, \overline{OE}=V_{IH}$, VPP must not be switched from VCC to VPP volts or vice versa.

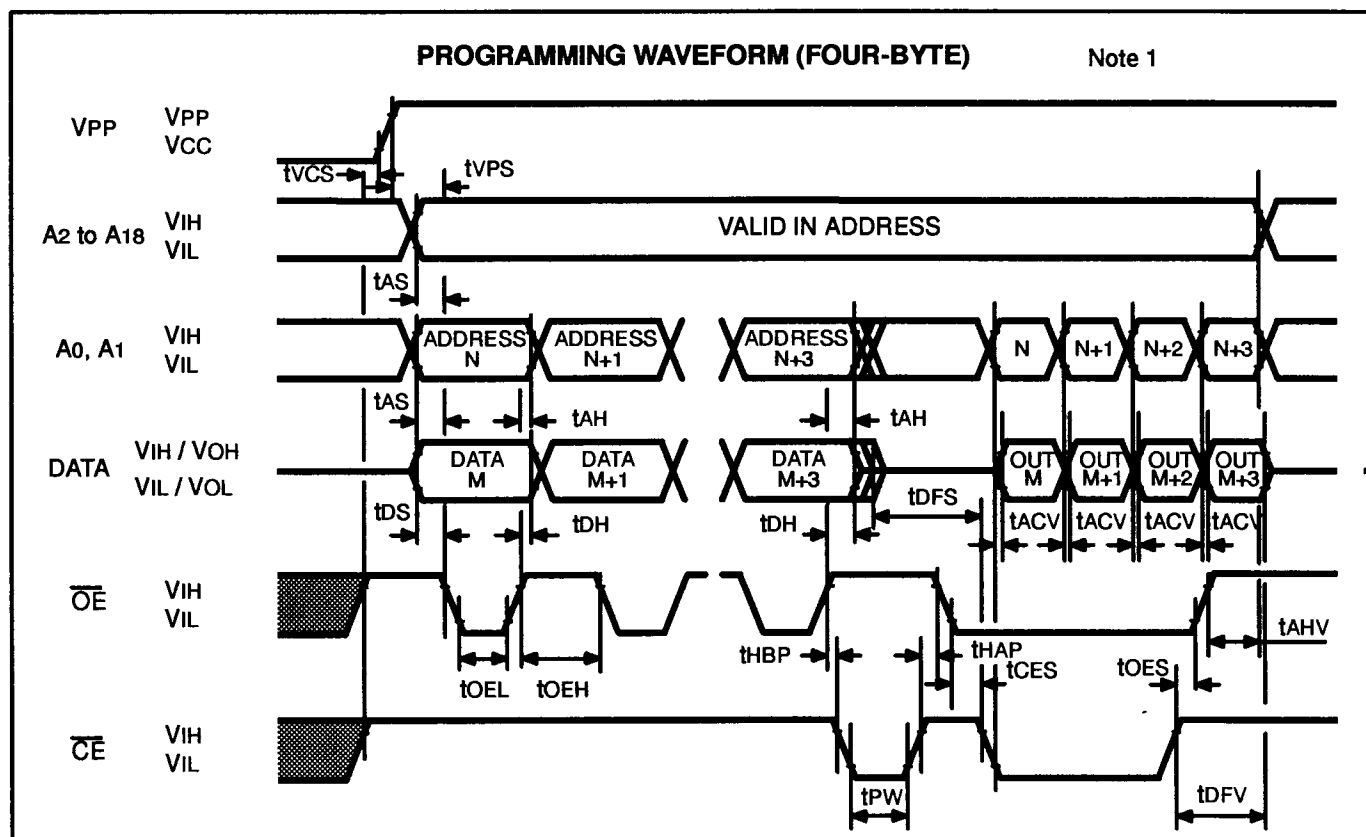
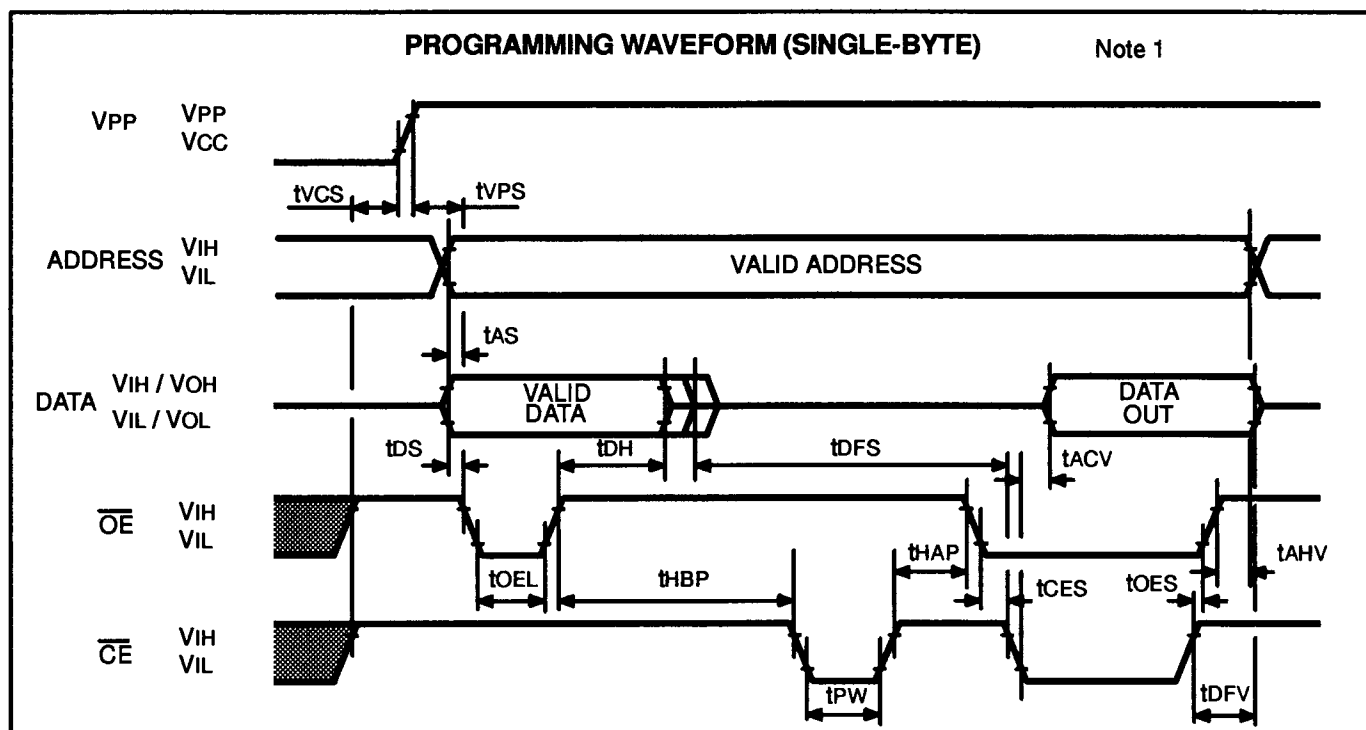
AC CHARACTERISTICS

(TA = 25 °C ± 5°C, VCC = 6V ± 0.25V, VPP = 12.5V ± 0.3V)

Parameter	Symbol	Values			Unit
		Min	Typ	Max	
Address Setup Time	t _{AS}	0.5			μs
Address Hold Time	t _{AH}	0.5			μs
Data Setup Time	t _{DS}	0.5			μs
Data Hold Time	t _{DH}	0.5			μs
$\overline{OE}$ Hold Time("L")	t _{OE L}	0.5			μs
$\overline{OE}$ Hold Time("H")	t _{OE H}	0.5			μs
Hold Time Before Programming	t _{HBP}	2			μs
Programming Pulse Width	t _{PW}	95	100	105	μs
Over Programming Pulse Number	N	1		50	times

AC CHARACTERISTICS(Cont'd)

Parameter	Symbol	Values			Unit
		Min	Typ	Max	
Hold Time After Programming	tHAP	2			μs
$\overline{\text{CE}}$ Setup Time	tCES	2			μs
$\overline{\text{OE}}$ Setup Time	tOES	2			μs
Input Data Floating Setup Time	tDFS	1			μs
Address Access Time at Verify	tACV			500	ns
$\overline{\text{OE}}$ to Output Float	tDFV			150	ns
Hold Time After Verify	tAHV	0			μs
VPP Setup Time	tVPS	20			μs
VPP Hold Time	tVCS	20			μs



Note 1: When verify fails, return to data latch.

PROGRAMMING / ERASING INFORMATION

PROGRAMMING FLOWCHART

Note 1 :

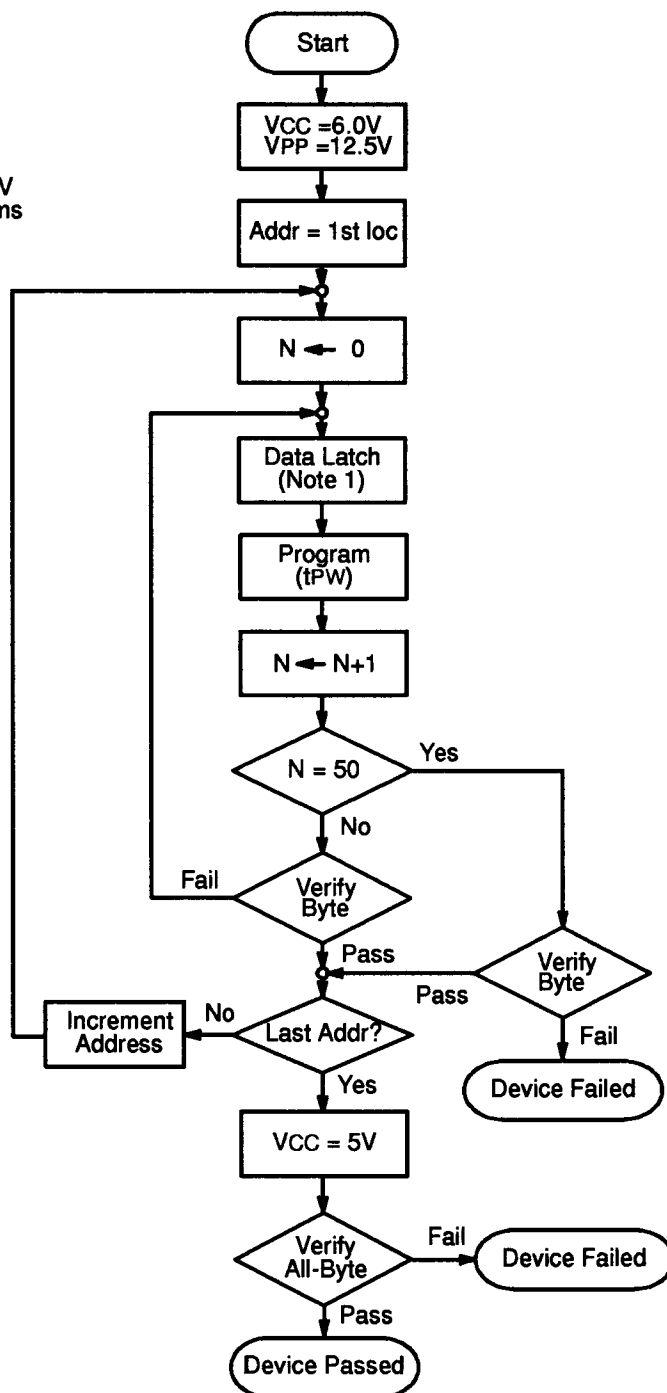
1. 1 byte or 4 bytes

2. Conditions:

$V_{CC} = 6(\pm 0.25)V$

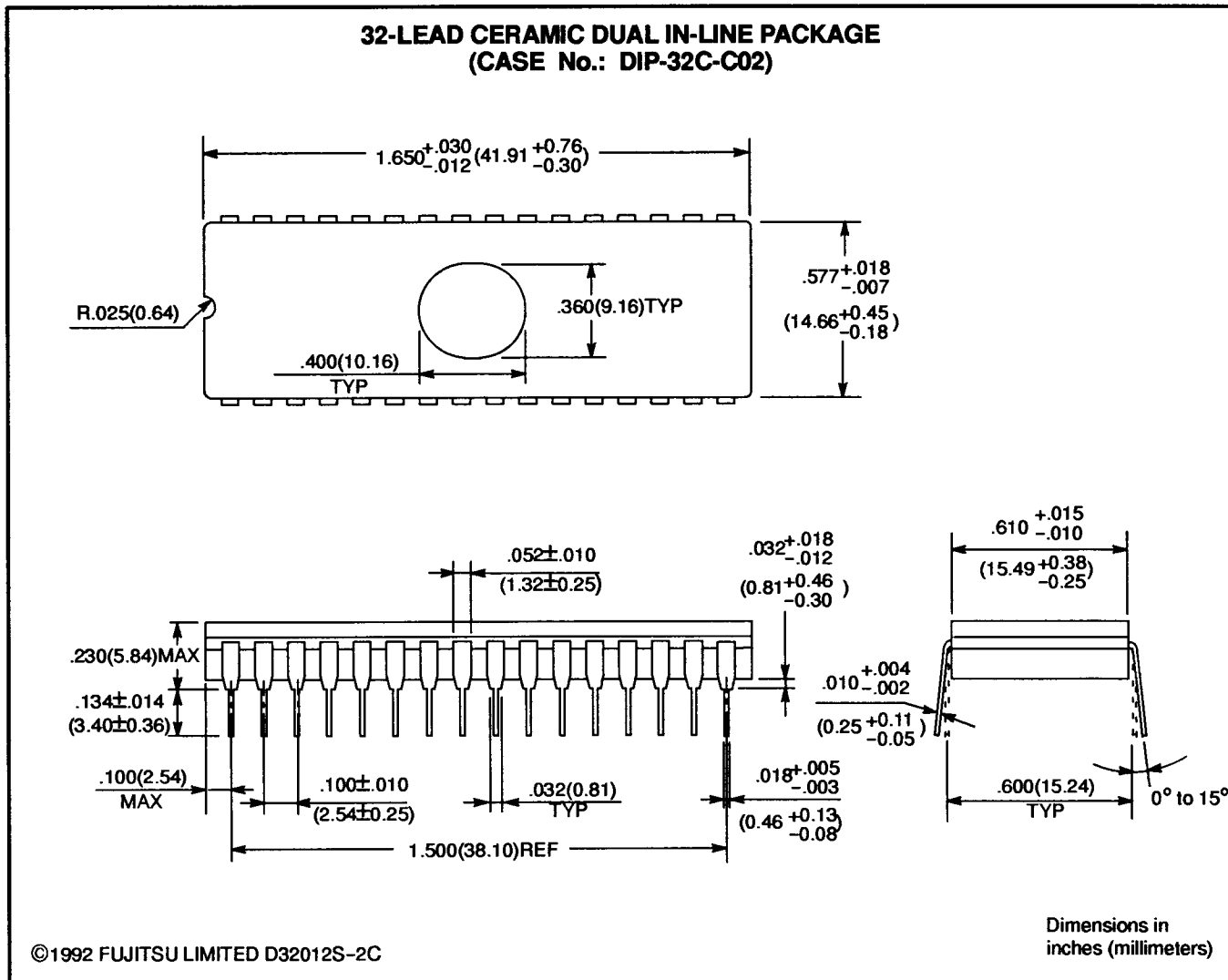
$V_{PP} = 12.5(\pm 0.3)V$

$t_{PW} = 0.1 \pm 0.005ms$



PACKAGE DIMENSIONS

(Suffix: Z)



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